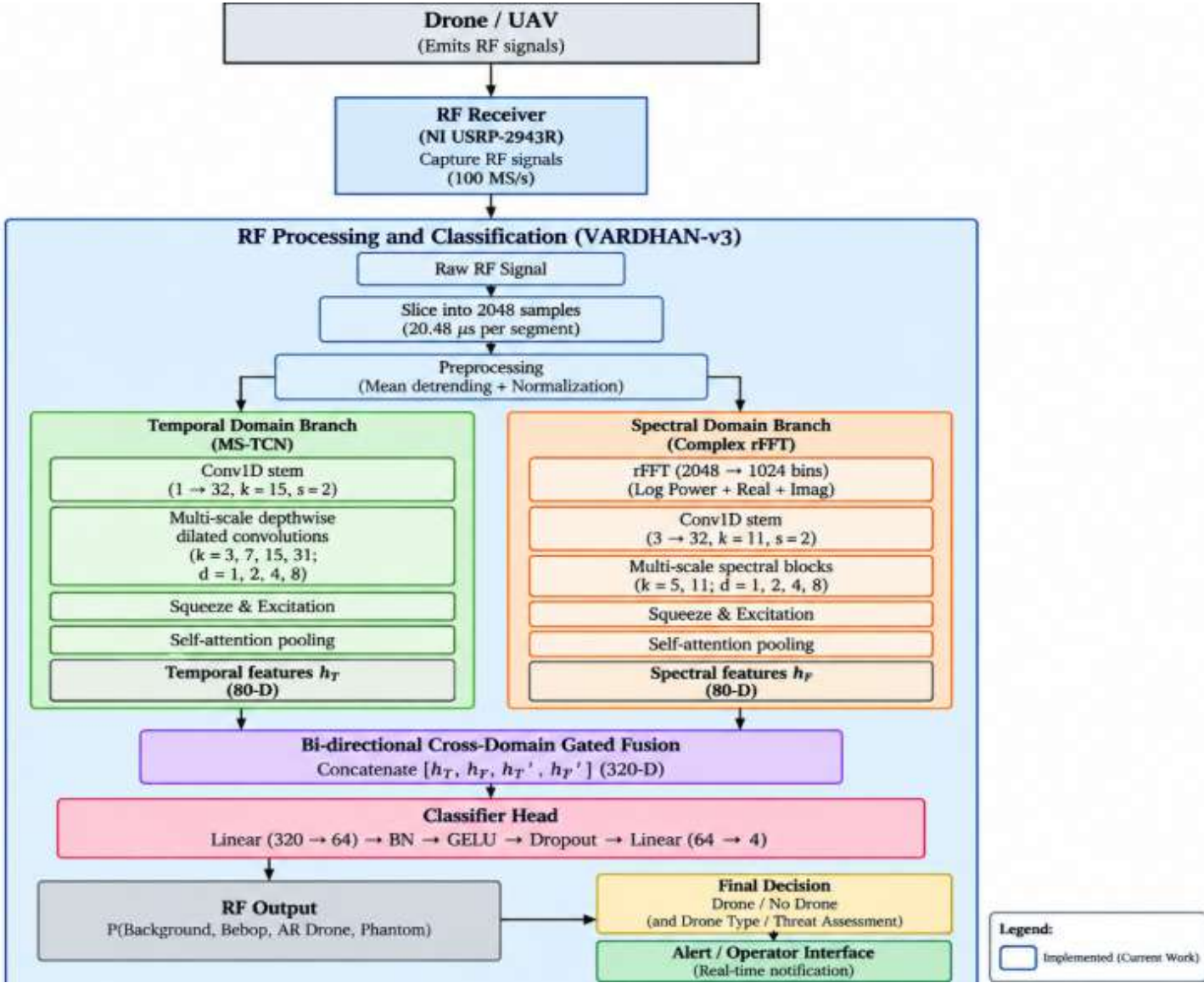
Drone / UAV
(Emits RF signals)
RF Receiver
(NI USRP-2943R)
Capture RF signals
(100 MS/s)
RF Processing and Classification (VARDHAN-v3)
Raw RF Signal
Slice into 2048 samples
(20.48 μs per segment)
Preprocessing
(Mean detrending + Normalization)
Temporal Domain Branch
(MS-TCN)
Conv1D stem
(1 → 32, k = 15, s = 2)
Multi-scale depthwise
dilated convolutions
(k = 3, 7, 15, 31;
d = 1, 2, 4, 8)
Squeeze & Excitation
Self-attention pooling
Temporal features h_T
(80-D)
Spectral Domain Branch
(Complex rFFT)
rFFT (2048 → 1024 bins)
(Log Power + Real + Imag)
Conv1D stem
(3 → 32, k = 11, s = 2)
Multi-scale spectral blocks
(k = 5, 11; d = 1, 2, 4, 8)
Squeeze & Excitation
Self-attention pooling
Spectral features h_F
(80-D)
Bi-directional Cross-Domain Gated Fusion
Concatenate $[h_T, h_F, h_T', h_F']$ (320-D)
Classifier Head
Linear (320 → 64) → BN → GELU → Dropout → Linear (64 → 4)
RF Output
P(Background, Bebop, AR Drone, Phantom)
Final Decision
Drone / No Drone
(and Drone Type / Threat Assessment)
Alert / Operator Interface
(Real-time notification)
Legend:
Implemented (Current Work)